

DECISION SUPPORT SYSTEMS CASE STUDY

Designing a DSS and GDSS for Traveloka

Regional decision-making in finance and human resources

Independent academic case study based on public information
Portfolio edition

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1. Introduction

Traveloka is one of the major travel platforms in Southeast Asia. It offers a wide range of services, including flights, accommodation, airport transfers, car rentals, attractions, cruises, villas, and apartments (Traveloka, n.d.). Because the company operates across different countries and manages many travel-related services at the same time, it needs to make decisions in areas such as pricing, budgeting, promotions, staffing, and customer service. These decisions can become difficult when managers have to work with large amounts of data while also coordinating with teams in different locations and time zones.

In this kind of business environment, a Decision Support System (DSS) can help managers make better decisions by providing useful data, forecasts, dashboards, and analytical tools. At the same time, a Group Decision Support System (GDSS) can help teams collaborate more effectively, especially when they are geographically dispersed. Research shows that GDSS is particularly useful when decision-makers are not in the same place and need structured tools for communication, discussion, and consensus building (Carneiro et al., 2021).

This report proposes a DSS and GDSS framework for Traveloka's regional decision-making. It first explains how a GDSS can be designed to improve teamwork and remote collaboration. It then discusses how the DSS can be implemented and managed in a practical way. After that, it examines how the DSS can support two important business areas, which are finance and human resources. Finally, it evaluates the ethical and social issues that may arise from using these systems. The report is written as a consultant-style proposal based on Traveloka's public business profile rather than a description of its internal non-public systems.

2. Company Background

Traveloka was founded in Indonesia in 2012 and later expanded into several regional markets, including Thailand, Vietnam, Singapore, Malaysia, Australia, and the Philippines. Public information also shows that Singapore is its global headquarters (Traveloka, n.d.; Traveloka Careers, n.d.). Besides transport and accommodation, the company also offers many other travel-related services. This means Traveloka must coordinate decisions across countries, customer groups, and product categories at the same time.

From a business perspective, a company like Traveloka is likely to face several common decision-making challenges. First, managers need timely and comparable performance

information across countries and product lines. Second, commercial teams need to respond quickly to changes in customer demand, which affects pricing and promotions. Third, HR managers need stronger planning tools for staffing and workforce allocation, especially during peak travel periods. Fourth, managers across countries need a better way to discuss, compare, and agree on decisions without being slowed down by fragmented communication or time-zone differences. Because of this, Traveloka is a suitable case for the use of both DSS and GDSS tools.

3. Task 1: Design of a Group Decision Support System (GDSS)

The GDSS for Traveloka should be designed to support collaboration between teams in different countries and functions. This includes executives, finance teams, HR managers, product teams, commercial teams, and customer-support leaders. According to Carneiro et al. (2021), modern GDSS should help decision-makers work together anytime and anywhere while reducing the communication problems that often happen in dispersed teams. For Traveloka, this means the system should not only support online meetings, but also improve how ideas are shared, evaluated, and finalized.

The GDSS should include three main groups of components: hardware, software, and communication tools. The hardware should include secure cloud servers, backup storage, laptops, desktops, mobile access devices, and meeting-room equipment such as microphones, cameras, and shared screens. These are important because regional managers need to access the system from different offices and sometimes remotely. The software should include dashboards, decision models, digital whiteboards, polling and voting tools, and scenario-analysis functions. These tools can help managers review the same data, compare options, and evaluate possible outcomes before making decisions. The communication tools should include video conferencing platforms, shared documents, enterprise messaging systems, and automated meeting-summary tools.

The GDSS should also support several decision-making processes. First, it should support brainstorming, where managers can share ideas for promotions, partnerships, service improvements, or market expansion. Second, it should help organize ideas into clear categories so that discussions do not become confusing. Third, it should support voting so teams can rank alternatives and choose priorities in a transparent way. Fourth, it should

support consensus building by ensuring that all participants can review the same evidence before a final decision is made. This is important because group decision-making research shows that diversity can improve decisions, but only when communication and coordination are well managed (Bang & Frith, 2017).

For Traveloka, the main value of the GDSS is that it can improve real-time and cross-border collaboration. Shared dashboards can make sure all teams are working with the same information. Voting tools can make prioritization more transparent and easier to justify. Digital records of discussions and decisions can improve accountability. At the same time, asynchronous participation functions can help teams contribute even when they are in different time zones. Structured discussion can also reduce the influence of hierarchy and encourage more balanced participation from regional teams. In this way, the GDSS would not only support online meetings, but also improve the overall quality of collective decision-making.

To make the GDSS more effective, its components should work together in a clear way. The hardware supports access and connectivity, the software supports structured decision processes, and the communication tools support interaction between regional teams. This integration is important because it allows the GDSS not only to support meetings, but also to improve decision quality by making information sharing, discussion, voting, and follow-up actions more systematic.

4. Task 2: Implementing and Managing the DSS

The DSS for Traveloka should be implemented in a phased and carefully managed way. The purpose of the system is not only to provide more reports, but to improve the overall quality, speed, and consistency of regional decision-making. More specifically, the DSS should help Traveloka unify decision-related data across markets, improve forecasting accuracy, support faster responses to market changes, and strengthen planning in finance and human resources. Broby (2022) explains that predictive analytics is becoming increasingly important in financial and managerial decision-making because it helps organizations use data not only to understand past performance, but also to forecast future outcomes.

Several stakeholders should be involved in the implementation process. Senior executives should define the strategic goals of the system and approve major decisions. IT and data teams should manage the technical infrastructure, system integration, and cybersecurity

controls. Finance managers should define dashboard needs, forecasting requirements, and budget-analysis functions. HR managers should identify workforce planning, performance monitoring, and staffing analysis needs. Country managers should ensure that the DSS reflects local business realities instead of only head-office priorities. Research on analytics implementation also shows that stakeholder alignment and organizational support are critical for making data-driven systems effective in practice (Wang et al., 2024).

The technical requirements of the DSS should be identified clearly from the beginning. In terms of hardware, Traveloka would need secure cloud servers, backup storage, analyst workstations, secure laptops for managers, and stable network access across regional offices. In terms of software, the company would need dashboard and business intelligence tools, database integration software, predictive analytics functions, role-based access controls, and reporting systems that can combine data from different sources. The main data sources should include booking data, revenue and payment records, campaign performance data, customer service data, HR records, payroll data, attendance records, and training or performance information. These requirements are important because the usefulness of the DSS depends on both data quality and system reliability.

Table 1:Implementation Timeline and Milestones

Month	Main activities	Milestone
Month 1	Complete requirement analysis, identify key users, and define system objectives.	Formal approval of project scope and business requirements.
Month 2	Finalize system architecture, select the platform or vendor, and confirm governance and security policies.	Architecture and vendor approval.
Month 3	Complete data integration and develop initial dashboards and reporting templates.	First working DSS prototype completed.
Month 4	Run a pilot test in selected functions or regional teams and collect user feedback.	Pilot completed and feedback reviewed.
Month 5	Roll out the DSS to more departments and country teams, supported by user training sessions.	Regional rollout completed.
Month 6	Conduct a post-implementation review, measure system performance, and make improvements.	First evaluation and optimization report completed.

After implementation, ongoing management is essential. Traveloka should establish a DSS governance committee responsible for data quality, access control, model review, system maintenance, and compliance. The company should also provide regular training so users understand how to use dashboards, how to interpret forecasts, and how to apply DSS outputs in real decisions. In addition, monthly data-quality checks, quarterly model reviews, and periodic user feedback sessions should be carried out to keep the system relevant and effective. Schilling et al. (2024) argue that DSS design should consider how much control remains with human users and how much is given to the system. This means the DSS should support managerial judgement rather than replace it.

5. Task 3: DSS Applications in Two Specific Domains

5.1 Finance

Finance is one of the most suitable areas for DSS use in Traveloka because the company needs to make many decisions about pricing, budgeting, campaign performance, and profitability across different markets. Since the travel industry is highly affected by seasonality and customer demand, managers need reliable forecasts and quick access to financial information. The DSS will be tailored to the finance function by focusing on revenue forecasting, budget allocation, profitability analysis, and campaign performance evaluation across different regional markets. Broby (2022) explains that predictive analytics in finance often uses methods such as regression, time-series models, and clustering to support better decision-making.

For Traveloka, a finance DSS should include revenue dashboards, budget-tracking tools, profitability analysis, demand forecasts, and what-if simulations. These tools could help managers compare performance across countries, estimate the likely effects of price changes, and decide where marketing budgets should be increased or reduced. Instead of relying only on past reports, managers could use the DSS to make more forward-looking and evidence-based decisions.

One practical example would be a quarterly regional budget review. Finance managers could compare booking performance, campaign returns, and profit margins across countries, then simulate the likely results of moving resources from one market to another. Another example would be planning promotions during major holiday periods. In that case, managers need to

know whether lower prices will increase sales enough to balance lower margins. A DSS can support these decisions by organizing the data clearly and showing possible outcomes before the final choice is made (Broby, 2022).

5.2 Human Resources

Human resources is another important area for DSS use because Traveloka's service quality depends strongly on its people. This includes customer-support teams, technical staff, product teams, and managers in different countries. HR decisions are especially important in a fast-moving business where demand can rise and fall depending on season, market, and promotions. The DSS will be tailored to the human resources function by focusing on workforce planning, staffing forecasts, attrition analysis, training needs, and employee performance monitoring across country operations.

An HR DSS for Traveloka should include workforce planning dashboards, staffing forecasts, attrition analysis, performance indicators, and training-needs analysis. These functions can help managers decide how many staff are needed, where turnover risk is rising, and which teams need more training. Wang et al. (2024) explain that HR analytics is most effective when it is connected to actual business decisions instead of being used only for administrative reporting. Tessema et al. (2025) also suggest that HR analytics can improve organizational performance when it supports strategic goals.

One example is seasonal staffing. If one market is expected to experience higher demand, the DSS could combine historical service data, booking forecasts, and current workforce information to show whether the company should hire more staff, reassign employees, or approve overtime. Another example is employee development. If one region has lower customer satisfaction, HR managers could use the DSS to see whether the problem is linked to training gaps, low staffing levels, or higher turnover. In this way, DSS can help HR managers make more proactive and better-informed decisions (Tessema et al., 2025; Wang et al., 2024).

6. Task 4: Ethical and Social Implications of DSS

Although DSS and GDSS can improve decision-making, they also create ethical and social concerns. One major issue is privacy. Traveloka's privacy notice states that the company may collect and manage personal data through its websites, apps, communications, and other

interactions, and that it uses technical and organizational protections to safeguard this information (Traveloka, 2025). If a DSS uses data connected to bookings, payments, communications, or employee records, then privacy protection must be treated as a key part of system design.

Another issue is fairness and bias. Research on AI governance highlights concerns such as transparency, fairness, explainability, and human oversight in data-driven systems (Papagiannidis et al., 2025). If the DSS uses biased historical data, it may produce recommendations that unfairly affect certain teams, markets, or employees. This is especially important in HR, where system outputs could influence staffing or performance-related decisions.

Transparency is also very important. If managers do not understand why the DSS recommends a certain budget adjustment, staffing level, or pricing action, they may either ignore the system or depend on it too heavily. Cheong (2024) argues that transparency and accountability are necessary because opaque decision systems can reduce trust. Schilling et al. (2024) also show that user-centered design matters because people need to feel that they still have control over important decisions.

There are also wider social effects. More data-driven decision-making may reduce some routine administrative work, but it may also create anxiety among employees who feel they are being monitored too closely or replaced by technology. Because of this, Traveloka should use a human-in-the-loop approach, where managers still make the final decision with support from the system. The company should also provide training, communicate clearly about the purpose of the DSS, and create a governance committee that includes legal, HR, data, and operations staff. These steps can help reduce risks and improve trust in the system (Cheong, 2024; Papagiannidis et al., 2025).

7. Conclusion

In conclusion, Traveloka is a suitable real-company example for a DSS and GDSS proposal because it operates across multiple markets and service categories, which creates a strong need for better regional decision-making. A GDSS could help managers in different countries work together more effectively through shared dashboards, structured discussion, voting tools, and consensus-building features. At the same time, a DSS could improve decision quality in

finance and human resources by providing forecasting, analysis, and scenario-planning tools (Broby, 2022; Carneiro et al., 2021).

However, successful implementation would depend on more than technology alone. Traveloka would also need clear governance, staff training, interpretable system outputs, and strong protections for privacy, fairness, and accountability. If these issues are managed properly, DSS and GDSS could improve not only decision speed, but also decision quality, teamwork, and regional alignment.

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